

Vishnu Patel

Python and Java Developer

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of hands-on experience designing and scaling Python-based and Java-based microservices in production environments. Strong expertise in Spring Boot, Kafka, and PostgreSQL, with a proven track record of improving system throughput, reliability, and deployment efficiency. Passionate about clean architecture, API-first design, and real-time data processing. Currently contributing to telecom-grade device management systems for smart utility networks, specifically managing DCU (Data Concentrator Units) and NIC (Network Interface Card) devices.

PROFESSIONAL SKILLS

Languages: Python, Java, SQL

Frameworks: Spring Boot, Spring Data JPA, REST APIs, Kafka Streams

Databases: PostgreSQL, MySQL

Messaging & Streaming: Apache Kafka

DevOps & Tools: Docker, Git, Postman, Swagger, PgAdmin, IntelliJ, STS

PROFESSIONAL EXPERIENCE (3+ Years)

COMMINT | Software Engineer

Nov 2022 – Present | Bengaluru, India

NMS Edge

Role: Developer | **Technologies:** Python, Flask, MQTT

- Improved device communication reliability by 90% using MQTT with efficient queuing, prioritization, and retries.
- Integrated MQTT with Flask APIs for low-latency, event-driven IoT communication and high-throughput data exchange
- Enabled message persistence and network resilience for robust remote monitoring and centralized device management.

FOTA Edge

Role: Developer | **Technologies:** Python, Flask, MQTT

- Developed an automated Flask-based system for remote firmware updates on electric meters, reducing manual effort and errors.
 - Used MQTT for reliable, real-time firmware delivery with optimized workflows and network efficiency.
- Implemented robust error-handling and a centralized monitoring system for real-time analytics and audit trails.

NMS Server (Device Service) Development

Domain: IoT & Network Management Systems

Role: Developer | **Technologies:** Java, Spring Boot, Kafka, PostgreSQL

- Developed and deployed high-performance Spring Boot microservices handling 50K+ daily device packets with <300ms latency.
- Integrated Apache Kafka for asynchronous event-driven communication, improving data reliability and scalability by 40%.
- Improved operational productivity by 25% through efficient API architecture.
- Optimized database queries and refactored code, boosting application performance by 20%.
- Designed and documented over 20+ RESTful APIs used by internal and external stakeholders, with a consistent API-first design.
- Built a real-time packet filtering engine using Kafka consumers that boosted data accuracy by 95%.
- Reduced database response time by 20% by optimizing PostgreSQL queries, indexing strategies, and pagination.

Serial Adapter Application

Role: Developer | **Technologies:** Java, Serial Communication, REST API, SIP, Spring Boot

- Implemented a serial communication adapter for hardware interoperability using Java and Spring Boot.
- Optimized low-bandwidth data transmission logic, enhancing reliability.
- Exposed a REST API to receive requests from external systems.
- Parsed REST requests and created SIP packets for device-level processing.

EDUCATION

Awadhesh Pratap Singh University, Rewa, Madhya Pradesh

Master of Computer Applications (MCA) 2019 - 2022

Percentage: 78%

Awadhesh Pratap Singh University, Rewa, Madhya Pradesh

Bachelor of Computer Applications (BCA)

2014 - 2017

Percentage: 69%

CERTIFICATIONS

- Full Stack Java Development Course - Cranes Varsity, Bengaluru (2022)
- SQL Certificate - HackerRank (2023)
- Java Certificate - HackerRank (2023)
- Java Multithreading, Concurrency & Performance Optimization - Udemy (2024)
- Building Microservices with Spring Boot & Spring Cloud - Udemy (2024)